

POWER BI 30 DAYS COURESE

Day 5 - Modeling Create Relationship

1. What is Modeling?

Modeling is the process of connecting different tables in Power BI.

It helps Power BI understand how data in one table is related to data in another table.

Example

Customers Table

Customer ID	Customer Name
101	Ram
102	Ravi

Orders Table

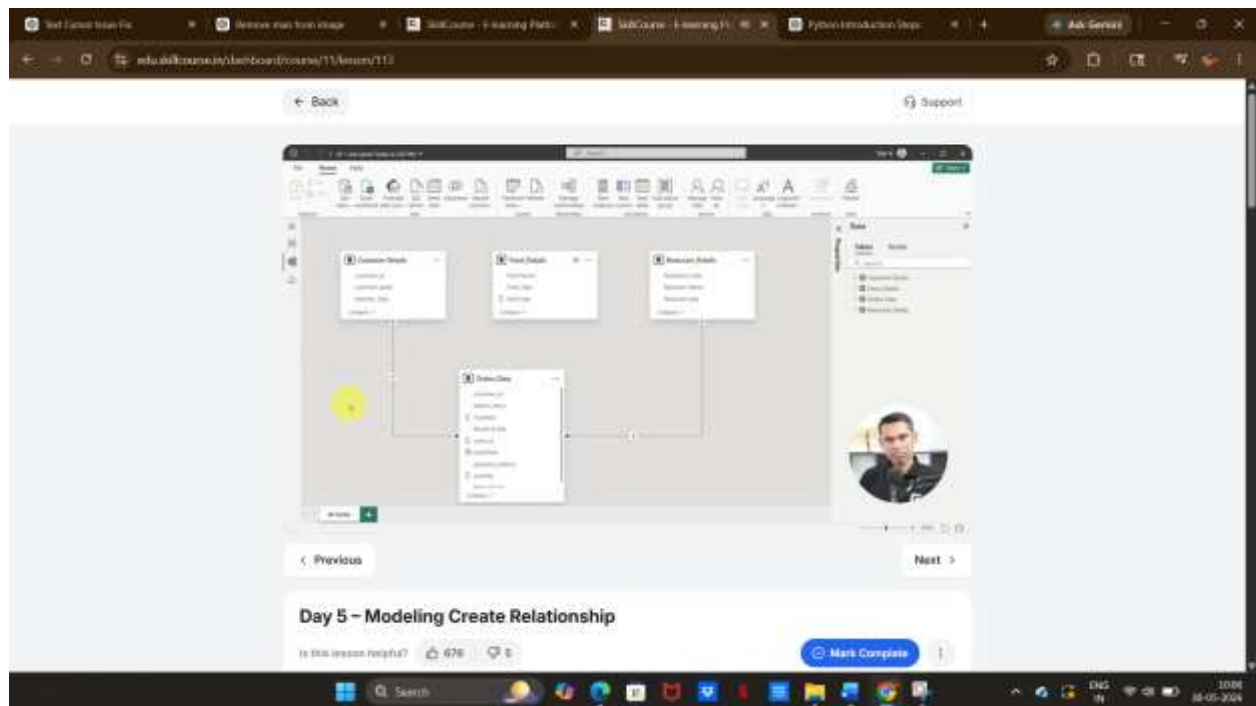
Order ID	Customer ID	Amount
1	101	500
2	102	1000

Here, Customer ID is common in both tables.

So we can connect both tables using Customer ID.

2. What is a Relationship?

A relationship is a connection between two tables using a common column.



Example

Customers[Customer ID]



Orders[Customer ID]

Now Power BI knows which customer placed which order.

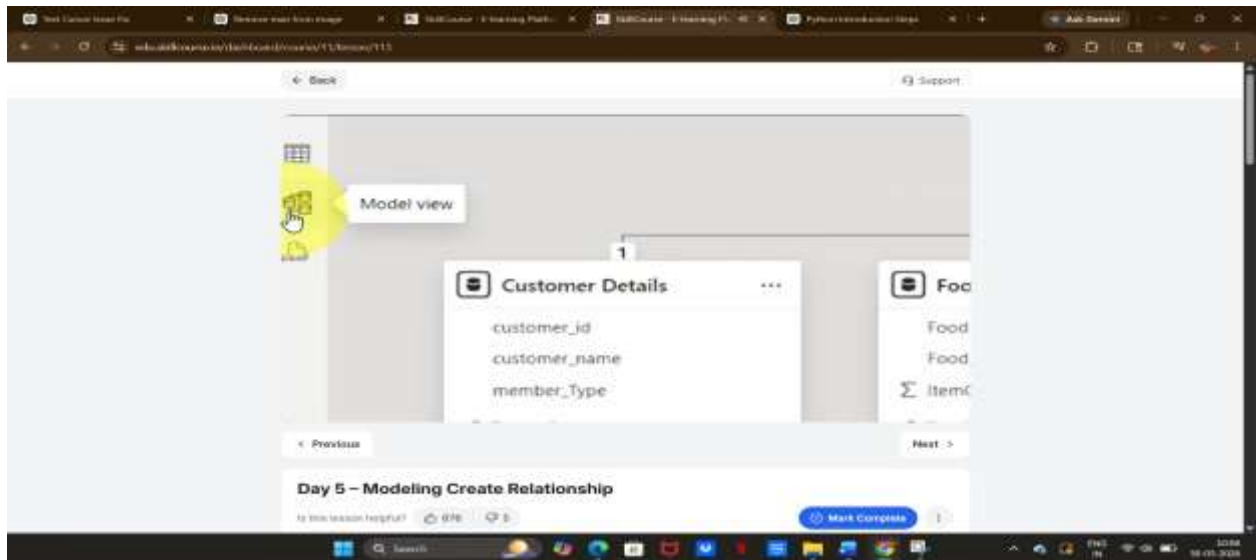
3. Why Do We Create Relationships?

Benefits

- Combine data from multiple tables
 - Avoid duplicate data
 - Create reports easily
 - Improve data analysis
 - Build dashboards correctly
-

4. Model View

Model View is used to see and manage relationships between tables.



Location

Left Side Panel → Model View

Use

- Create relationships
- Edit relationships
- Delete relationships
- View table connections

5. Primary Key A Primary Key is a column that contains unique values.

Example

Customer ID
101
102
103

No duplicates.

Use

Used on the One side of a relationship.

6. Foreign Key

A Foreign Key is a column that contains repeated values.

Example

Customer ID
101
101
102
103

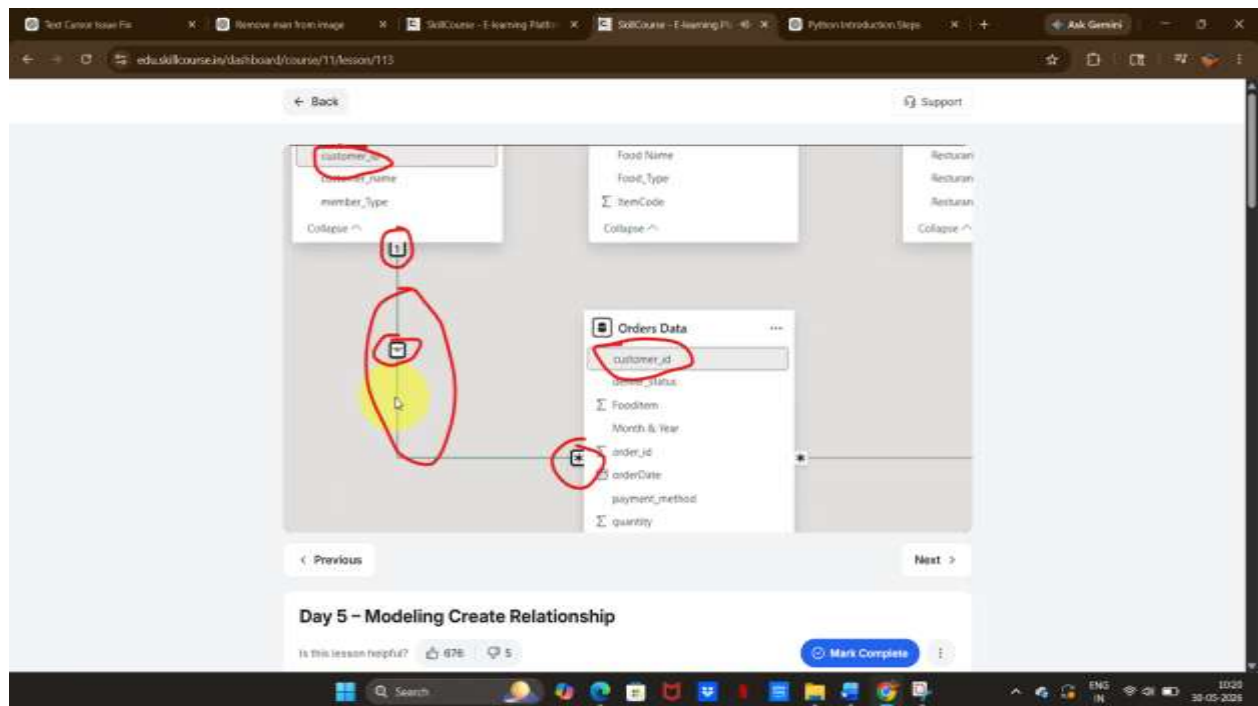
Duplicates are allowed.

Use

Used on the Many side of a relationship.

7. Cardinality

Cardinality defines how records in one table are related to records in another table.



A. One-to-Many (1:*)

Most commonly used.

Example

Customer Table

Customer ID
101
102

Orders Table

Customer ID
101
101
102

One customer can place many orders.

Symbol

1 → *

B. Many-to-One (*:1)

Reverse of One-to-Many.

Many orders belong to one customer.

Symbol

• → 1

C. One-to-One (1:1) means

One record in Table A matches exactly one record in Table B.

Example

Employee Table

Employee ID	Name
101	Ram

Employee Details Table

Employee ID	PAN Number
101	ABC123

Here:

- Employee ID 101 appears only once in Employee Table.
- Employee ID 101 appears only once in Employee Details Table.

One record ↔ One record

D. Many-to-Many (:) means

Many records in one table can match many records in another table.

Example

Students Table

Student
Ram
Ravi

Subjects Table

Subject
Math
Science

Relationships:

- Ram studies Math and Science
- Ravi studies Math and Science

So:

- One Student → Many Subjects
- One Subject → Many Students

Many ↔ Many

Filters in Power BI

Easy Meaning

Filters are used to display only the required data and hide unwanted data.

Example

Suppose you have sales data from many countries:

Country	Sales
India	5000
USA	7000
UK	3000

If you apply a filter for India,

Result:

Country	Sales
India	5000

Only India's data will be shown.

Why Use Filters?

- Show specific data
 - Analyze data easily
 - Hide unnecessary records
 - Create focused reports
-

Types of Filters in Power BI

1. Visual Level Filter

Affects only one visual/chart.

2. Page Level Filter

Affects all visuals on the current page.

3. Report Level Filter

Affects all pages and visuals in the report.

Easy Interview Answer

Filters are used to show only the required data in a report by hiding unwanted data. They help users analyze specific information quickly and easily.

8. Cross Filter Direction

It decides which table can affect the other table when a filter is applied.

A. Single Direction

Filter moves in one direction only.

Example

Customer Table

Customer ID	Customer Name
101	Ram
102	Ravi

Orders Table

Customer ID	Sales
101	500
101	1000
102	700

If you select Ram in Customer Table,
Power BI will show only Ram's orders.

Customer filters Orders

Orders cannot filter Customer

Easy Definition

Single Direction means the filter moves in only one direction from one table to another.

B. Both Direction

Filter moves in both directions.

Example

Customer ↔ Orders

If Customer is filtered → Orders change

If Orders are filtered → Customer changes

Easy Definition

Both Direction means tables can filter each other.

9./10. Active and Inactive Relationship (Based on Your Image)

Inactive Relationship (Red Dotted Line)

Ship Date → Calendar Date

✗ This relationship is inactive.

Power BI does not use it automatically.

Example

If you create a chart:

Sales by Month

Power BI will still use Order Date, not Ship Date.

To use Ship Date, you must explicitly tell Power BI using DAX (USERELATIONSHIP function).

Why is Ship Date Inactive?

Because both Order Date and Ship Date are connected to the same Calendar table.

Power BI allows only one active relationship between the same two tables to avoid confusion.

If both were active, Power BI would not know:

- Should sales be shown by Order Date?
- Or by Ship Date?

Therefore:

☒ Order Date → Active (Solid Line)

✗ Ship Date → Inactive (Dotted Line)

Easy Interview Answer

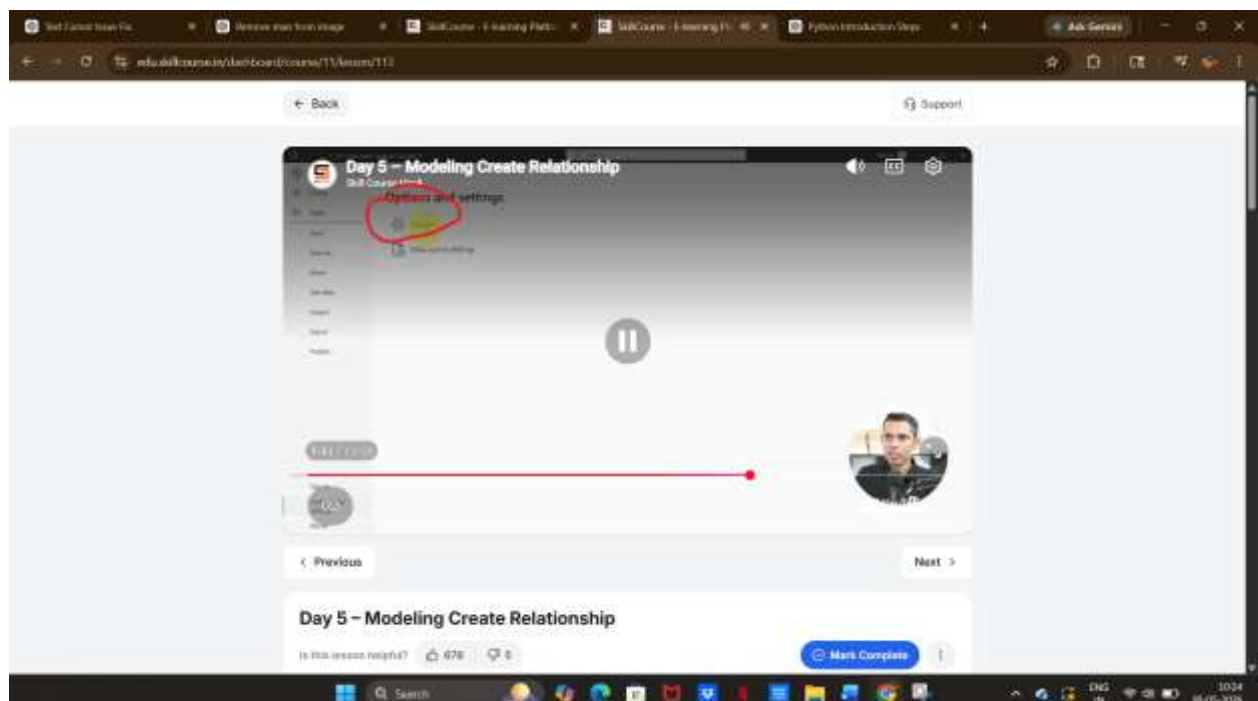
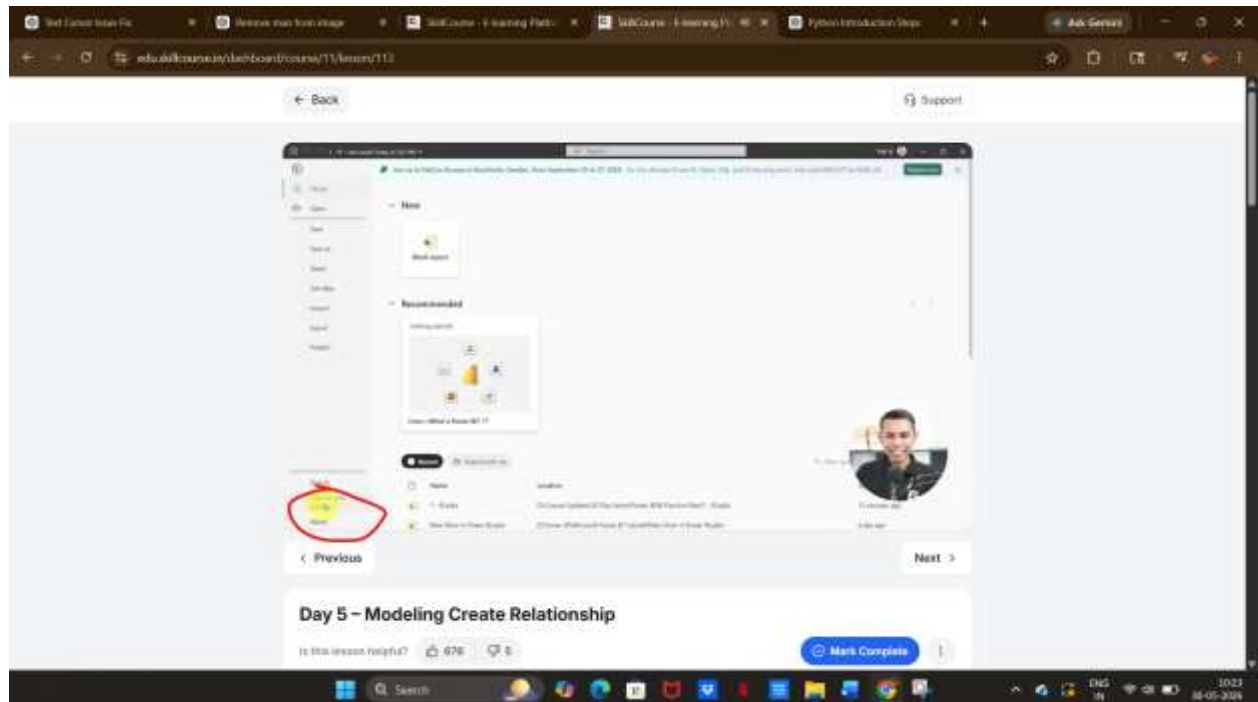
An Active Relationship is used automatically by Power BI and is shown by a solid line. An Inactive Relationship exists in the model but is not used automatically and is shown by a dotted line. In the image, Order Date is active and Ship Date

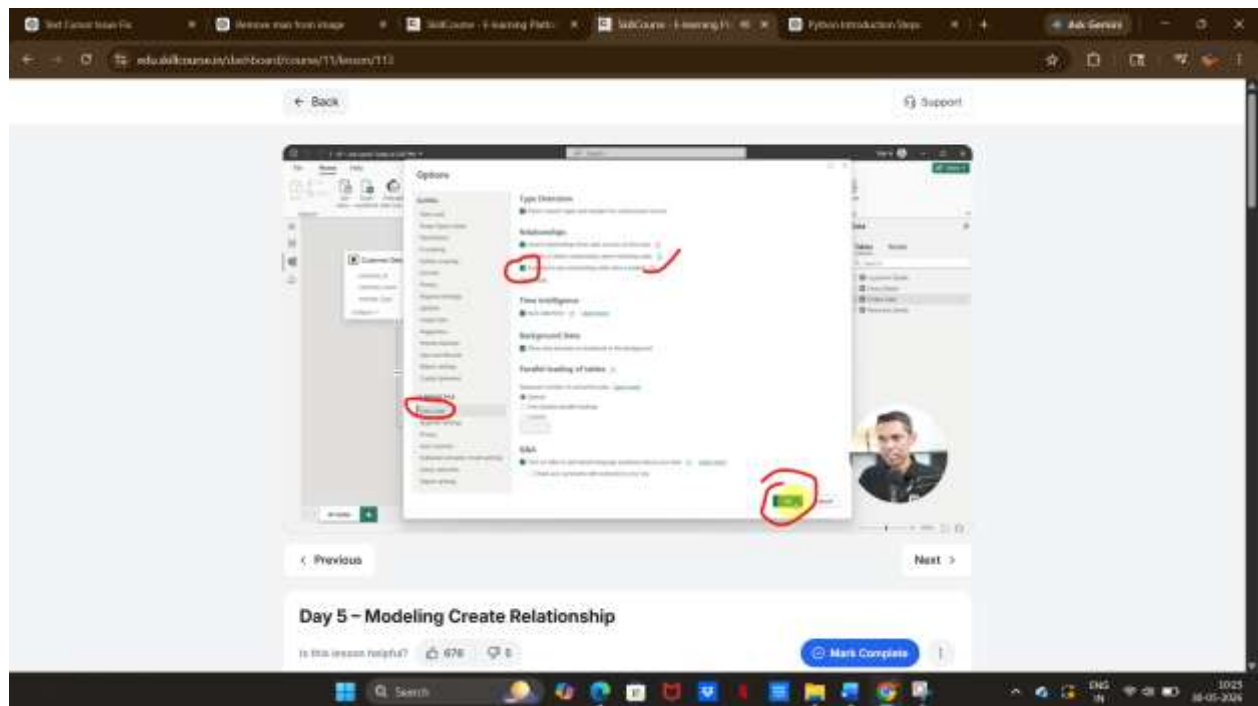
is inactive because both connect to the same Calendar table.

11. Auto Detect Relationship

Power BI can automatically create relationships if common columns are found.

Sometimes power bi does not create auto relationship, to make it auto go options and .. → options → select autodetect new relationship...





Example

Table 1

Customer ID
101
102

Table 2

Customer ID
101
102

Power BI detects the common column and creates the relationship automatically.

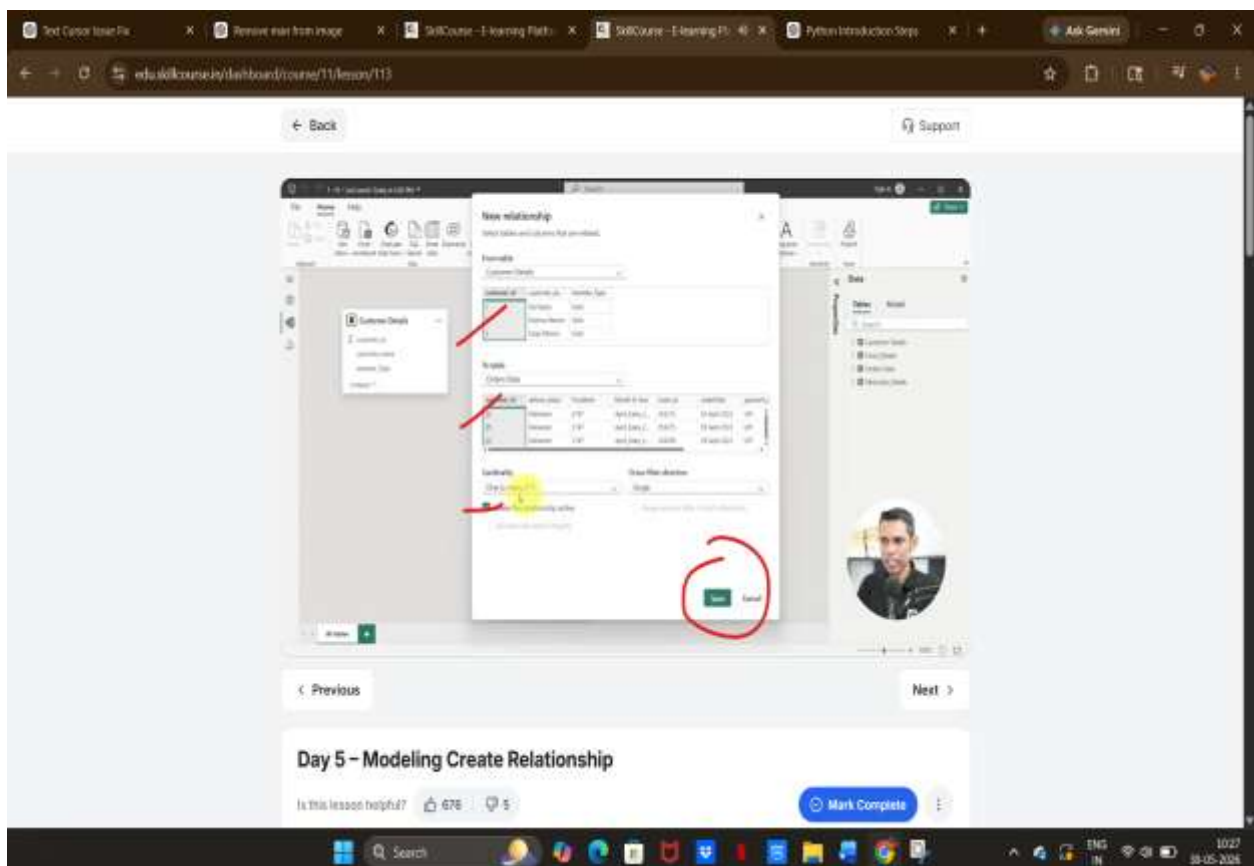
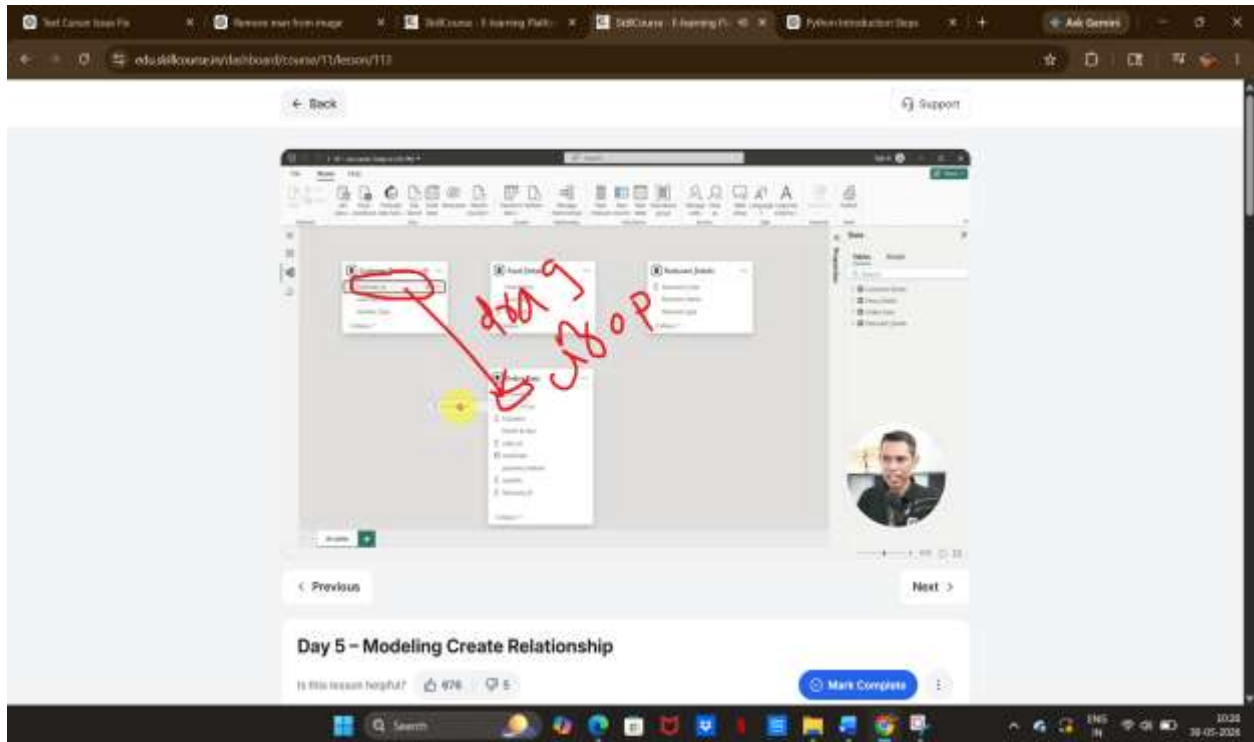
Easy Definition

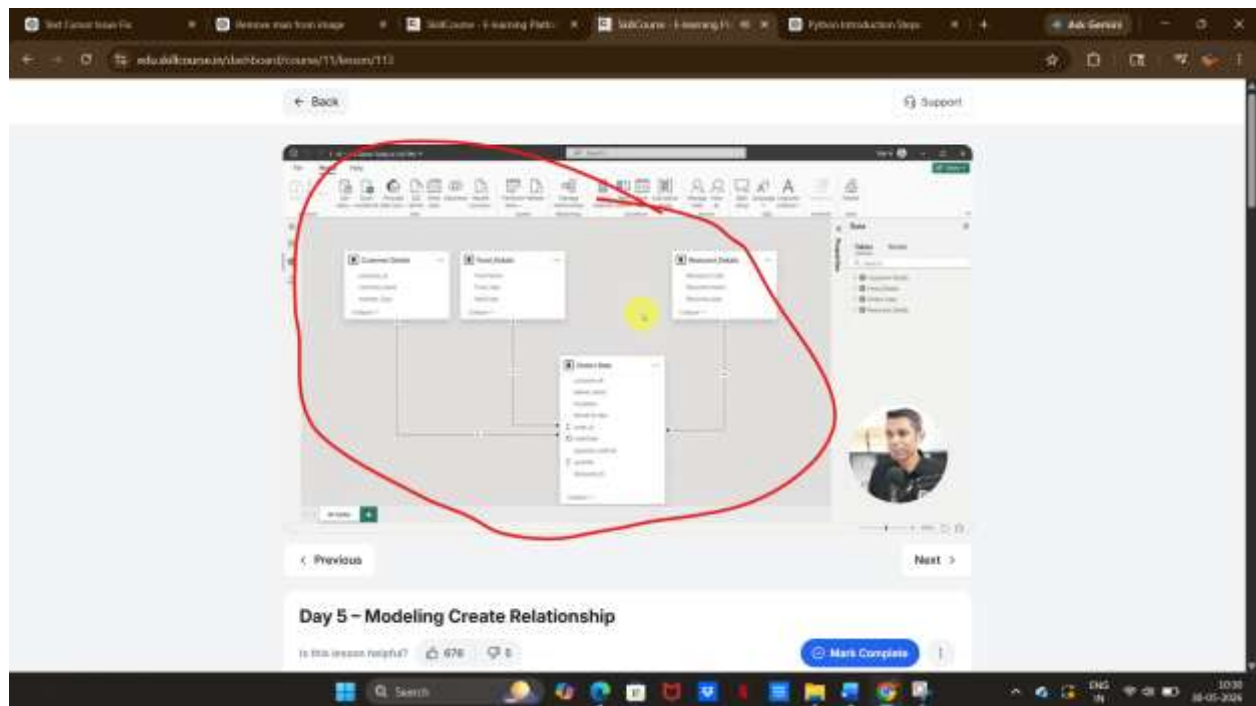
Auto Detect automatically creates relationships between tables having common columns.

12. Create Relationship Manually

Sometimes Power BI cannot detect relationships.

Then we create them manually.





Steps

Step 1: Open Model View

Step 2: Select the common column

Step 3: Drag it to the matching column in another table

Step 4: Relationship window opens

Step 5: Select:

- Cardinality
- Cross Filter Direction

Step 6: Click OK

Easy Definition

Manual Relationship means connecting tables yourself using a common column.

13. Edit Relationship

If you want to change an existing relationship.

Steps

Home → Manage Relationships → Edit

You can change:

- Cardinality
- Cross Filter Direction
- Active/Inactive status

Easy Definition

Edit Relationship is used to modify an existing relationship.

14. Delete Relationship

Used when a relationship is no longer needed.

Steps

Right Click Relationship Line → Delete

Easy Definition

Delete Relationship removes the connection between two tables.

15. Star Schema

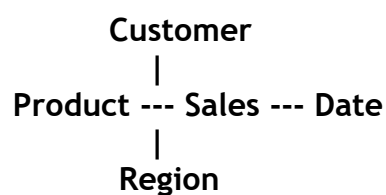
Star Schema is the best and most commonly used data model in Power BI.

Structure

One Fact Table in the center

Connected to multiple Dimension Tables

Example



Looks like a ★ star.

Benefits

- Faster reports

- Better performance
- Easy to understand
- Easy to manage

Easy Definition

Star Schema is a model where one Fact Table is connected to multiple Dimension Tables.

Fact Table vs Dimension Table (Unique & Duplicate Values)

Dimension Table

Dimension tables usually contain Unique Values in their key column.

Example: Customer Details

Customer ID	Customer Name
101	Ram
102	Ravi
103	Amit

Here:

☒ Customer ID is Unique

☒ No duplicates

This is called the One Side (1) of the relationship.

Fact Table

Fact tables usually contain Duplicate Values in their foreign key columns.

Example: Orders Data

Order ID	Customer ID	Sales
1	101	500
2	101	1000
3	102	700

Here:

Customer ID repeats.

☒ Duplicates are allowed

This is called the Many Side (*) of the relationship.

Relationship Example

Customer Details (Dimension)

Customer ID
101
102
103

Orders Data (Fact)

Customer ID
101
101
102

Customer ID
103
103

Relationship:

Customer Details (1) ----- (*) Orders Data

Meaning

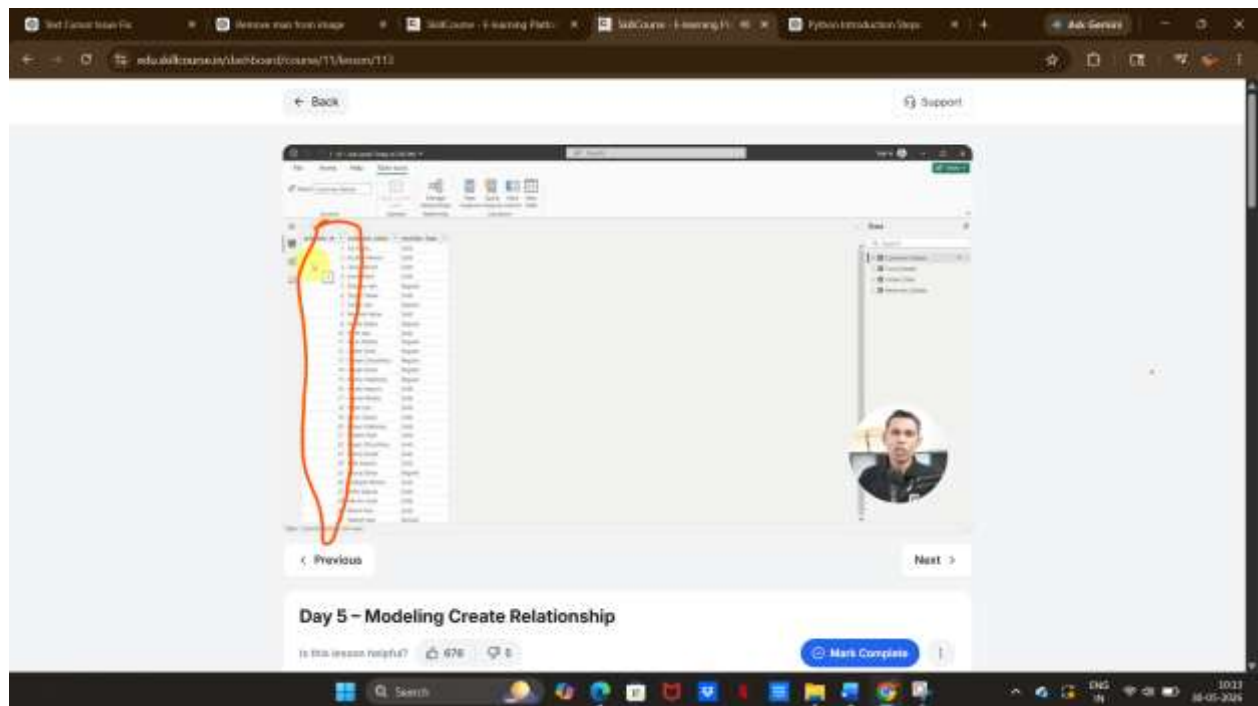
- One Customer → Many Orders
- Unique Values → Dimension Table
- Duplicate Values → Fact Table

The screenshot shows a video lecture interface with a browser window displaying a Power BI relationship diagram. The diagram illustrates a one-to-many relationship between 'Customer Details' and 'Orders Data'. Handwritten orange annotations are present: a red box around the 'Customer Details' table is labeled 'Dimension table', and a red box around the 'Orders Data' table is labeled 'Fact table or Data table'. The video title at the bottom is 'Day 5 - Modeling Create Relationship'. The Windows taskbar at the bottom shows the date as 10-05-2024.

Web browser tabs: Test 7: your team file, Review your team stage, SkillCourse - Learning Paths, SkillCourse - Learning Paths, Python Introduction Steps, Ask Google.

Address bar: edureka.com/learn/course/11/lesson/113

Navigation: Back, Support



Day 5 - Modeling Create Relationship

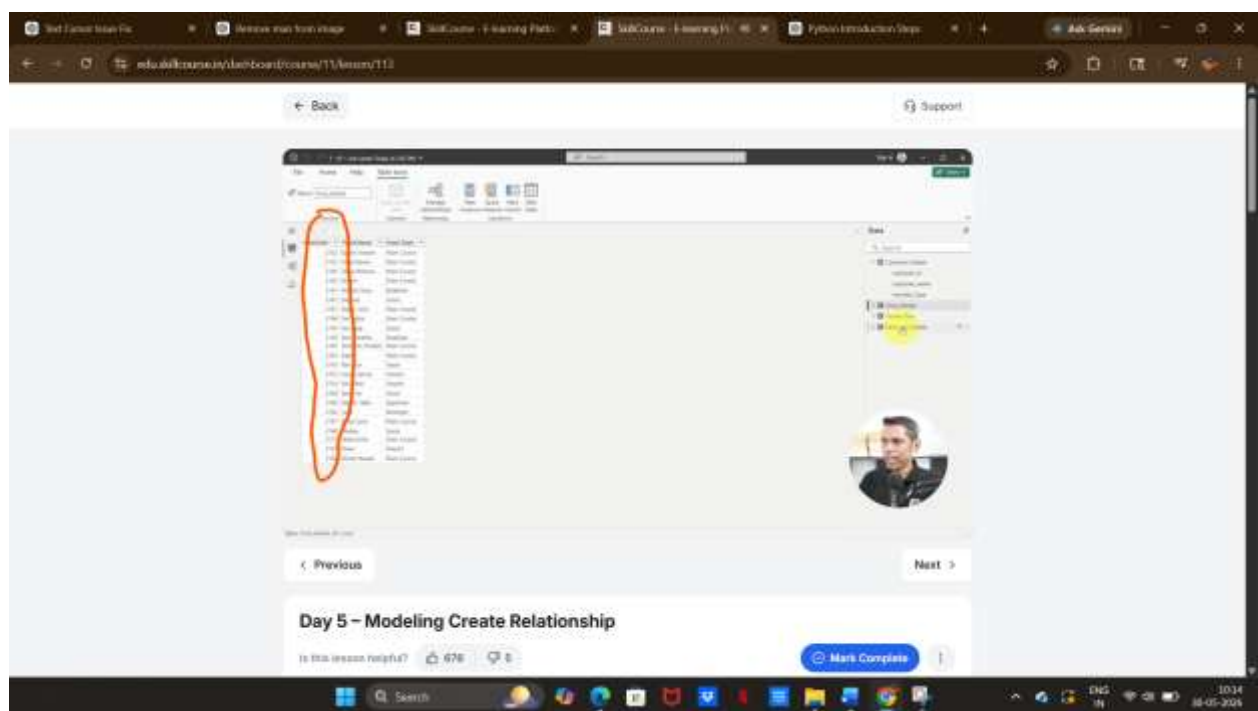
Is this lesson helpful? 678

Mark Complete

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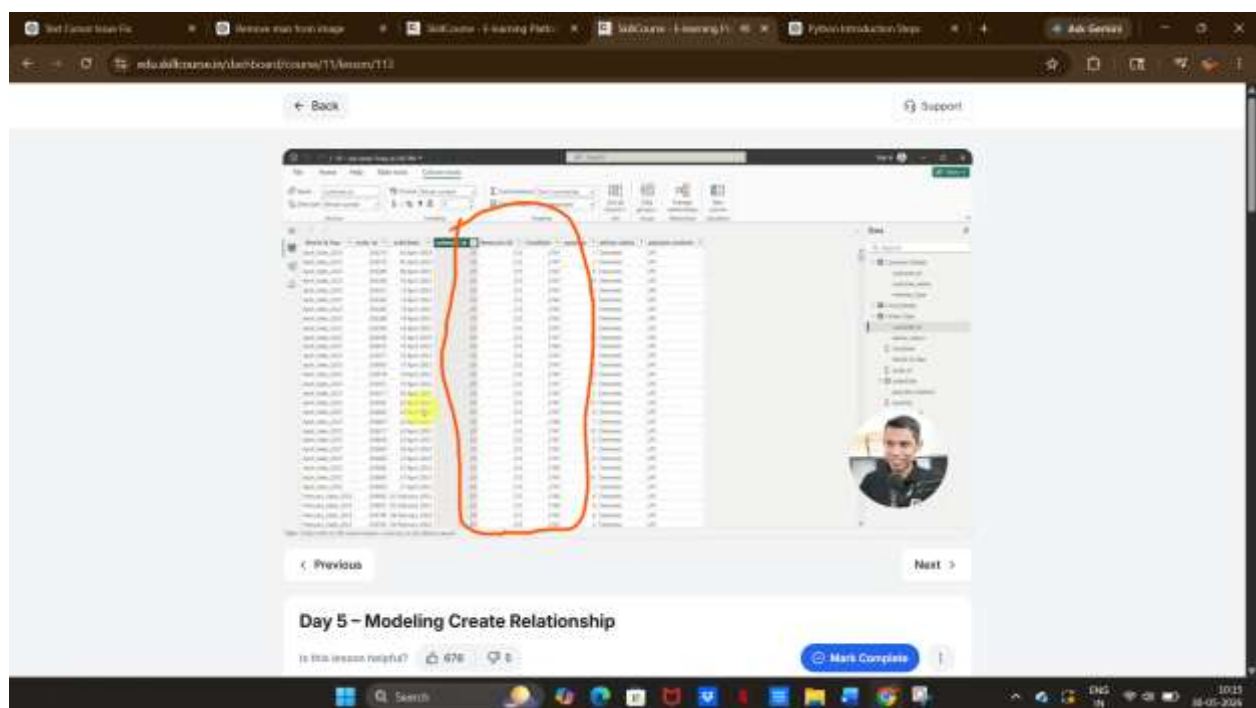
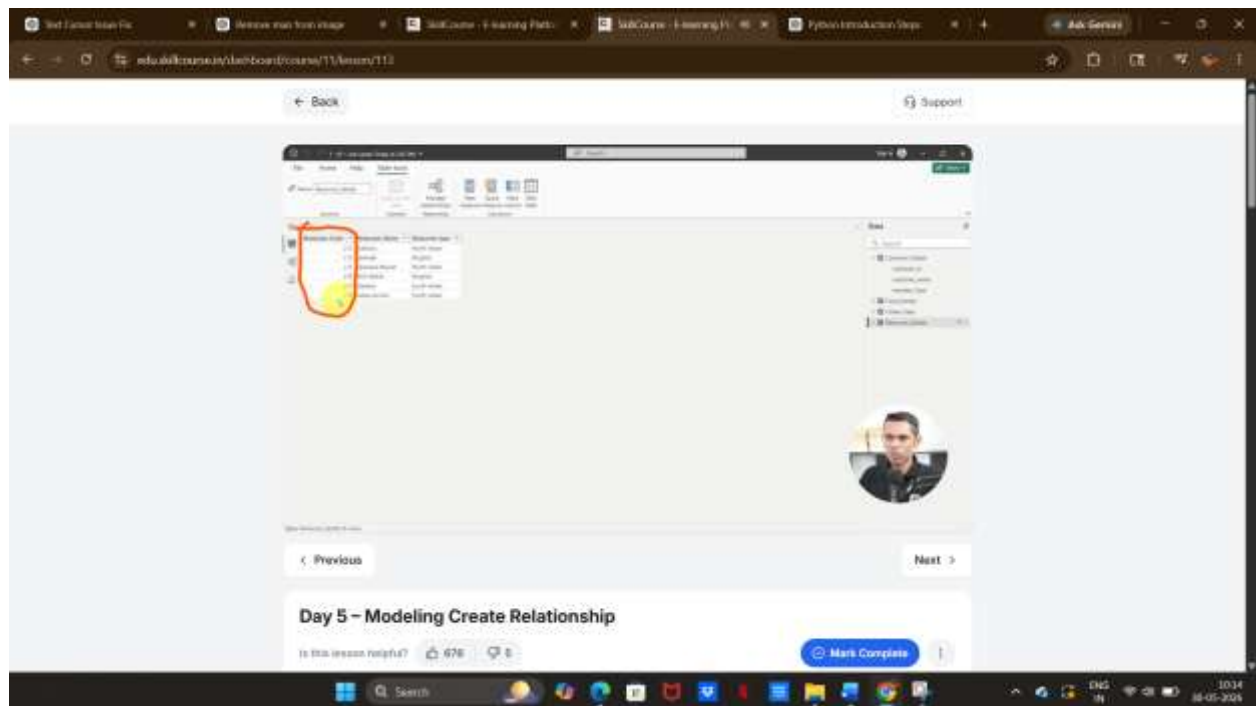
Navigation: Back, Support



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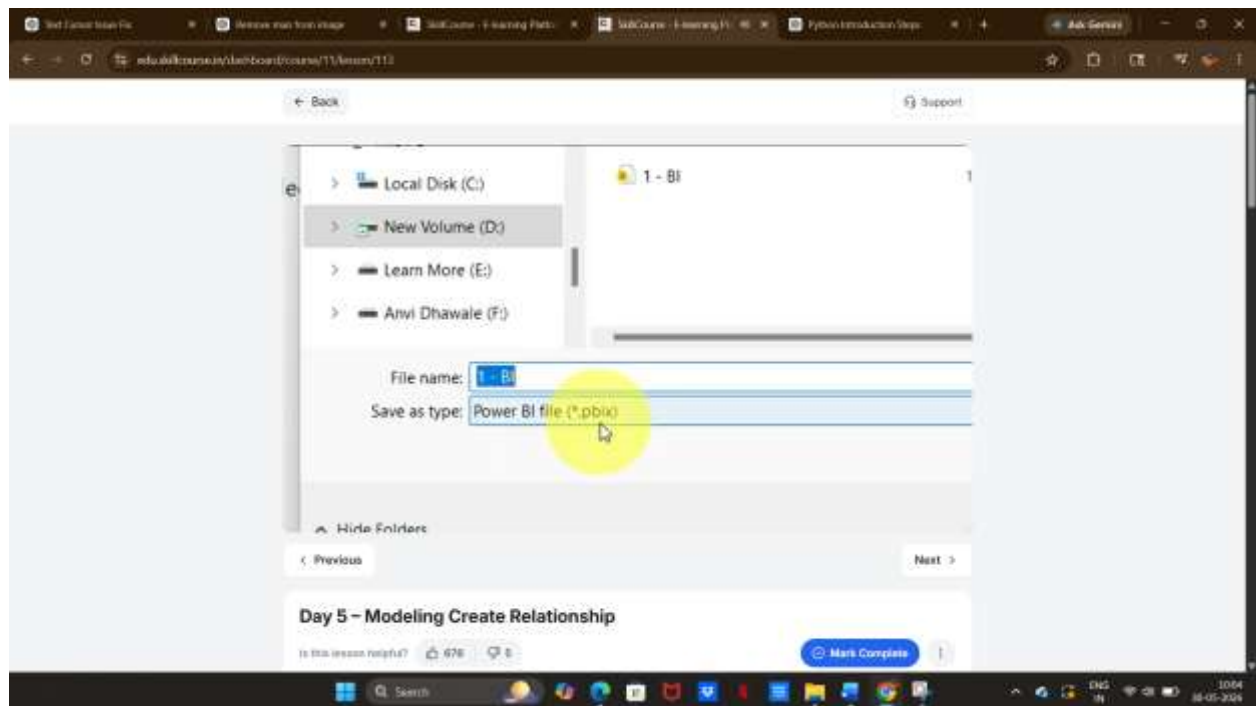
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Easy Interview Answer

Dimension Tables contain unique values and descriptive information such as Customer Name and Product Name. Fact Tables contain duplicate values and transactional data such as Sales, Profit, and Quantity. In a relationship, the Dimension Table is on the One (1) side and the Fact Table is on the Many (*) side. 🚀



Save Power BI File

Steps

1. Click **File**
2. Select **Save As**
3. Choose a folder location
4. Enter the file name
5. Click **Save**

Power BI File Extension

.pbix

Example

Sales Dashboard.pbix

or

Restaurant Analysis.pbix

What is a .pbix File?

A .pbix file is the default Power BI Desktop file format.

It stores:

- Data
- Relationships
- Queries
- DAX Measures
- Reports
- Dashboards

in a single file.

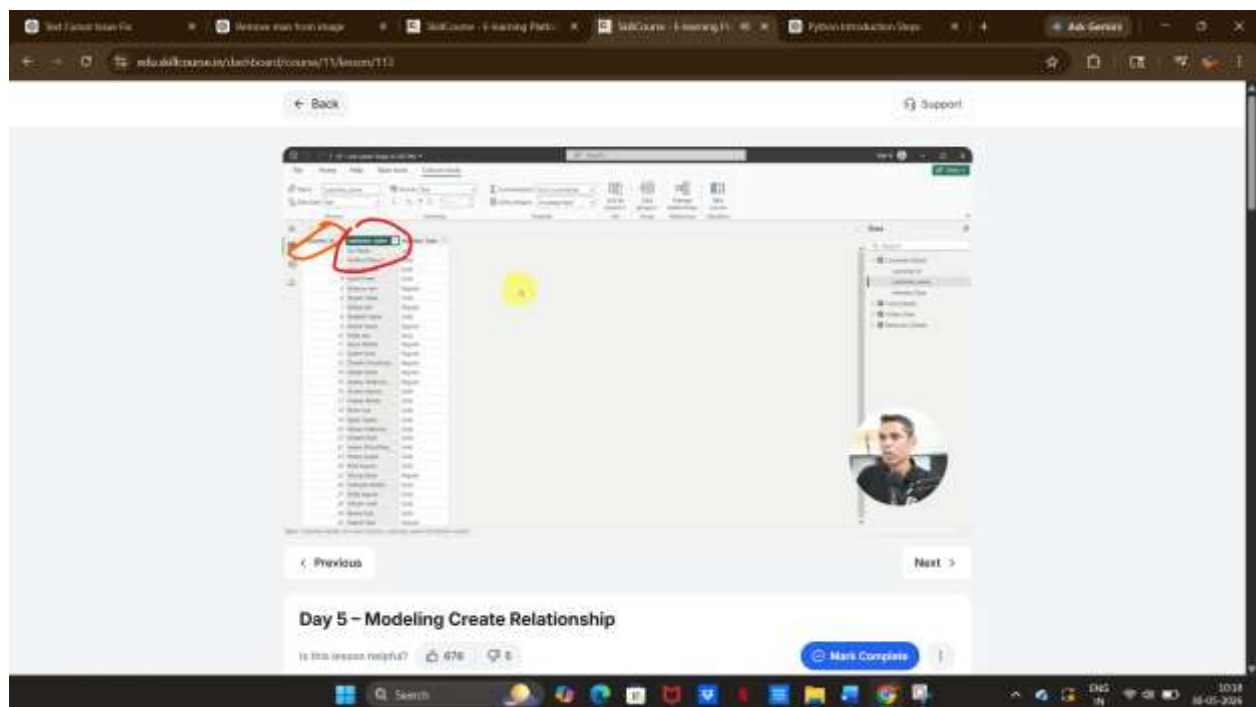
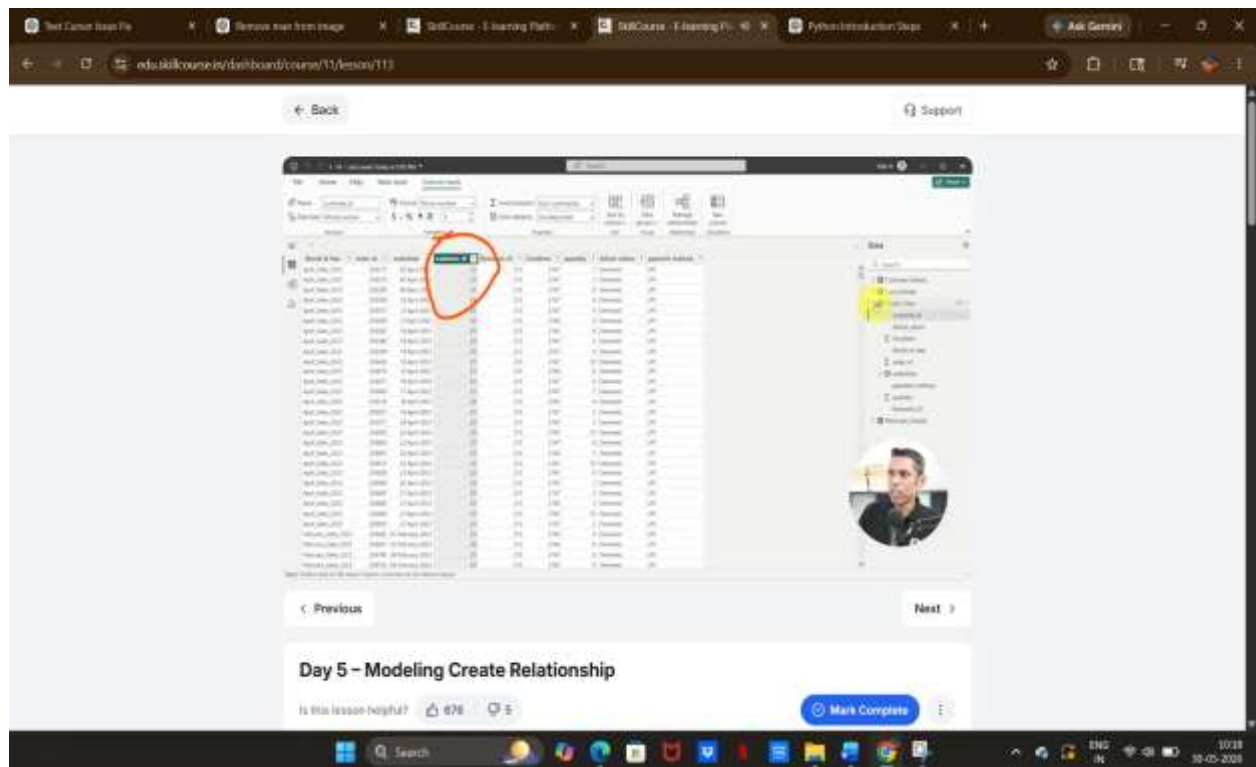
Why Save the File?

To:

- Avoid losing work
 - Reopen the project later
 - Share the project with others
 - Continue report development
-

Now, we will create relationship btw two tables

Step 1: we will create relationship because in order table we do not have customer name but customer_detail table has customer_name so to know name of customer we need to make relationship order data table has customer_id and customer_detail table has also customer_id so because of same column we can make same relationship.



Step 2: we will drag customer_id from customer_detail table to order data table
 Here we have one to many relationship means one customer can order multiple orders.

Test Canvas Issue Fix x Remove man from image x SkillCourse - E-learning Platform x SkillCourse - E-learning Platform x Python Introduction Steps x Ask Gemini

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Back Support

Customer Data

- customer_id
- customer_name
- member_type

Food Data

- Food Name
- Food Type
- Item Code

Orders Data

- customer_id
- order_status
- Food Item
- Month & Year
- order_id
- order Date
- payment_method
- quantity

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Mark Complete

Windows Search

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11:27 / 13:58

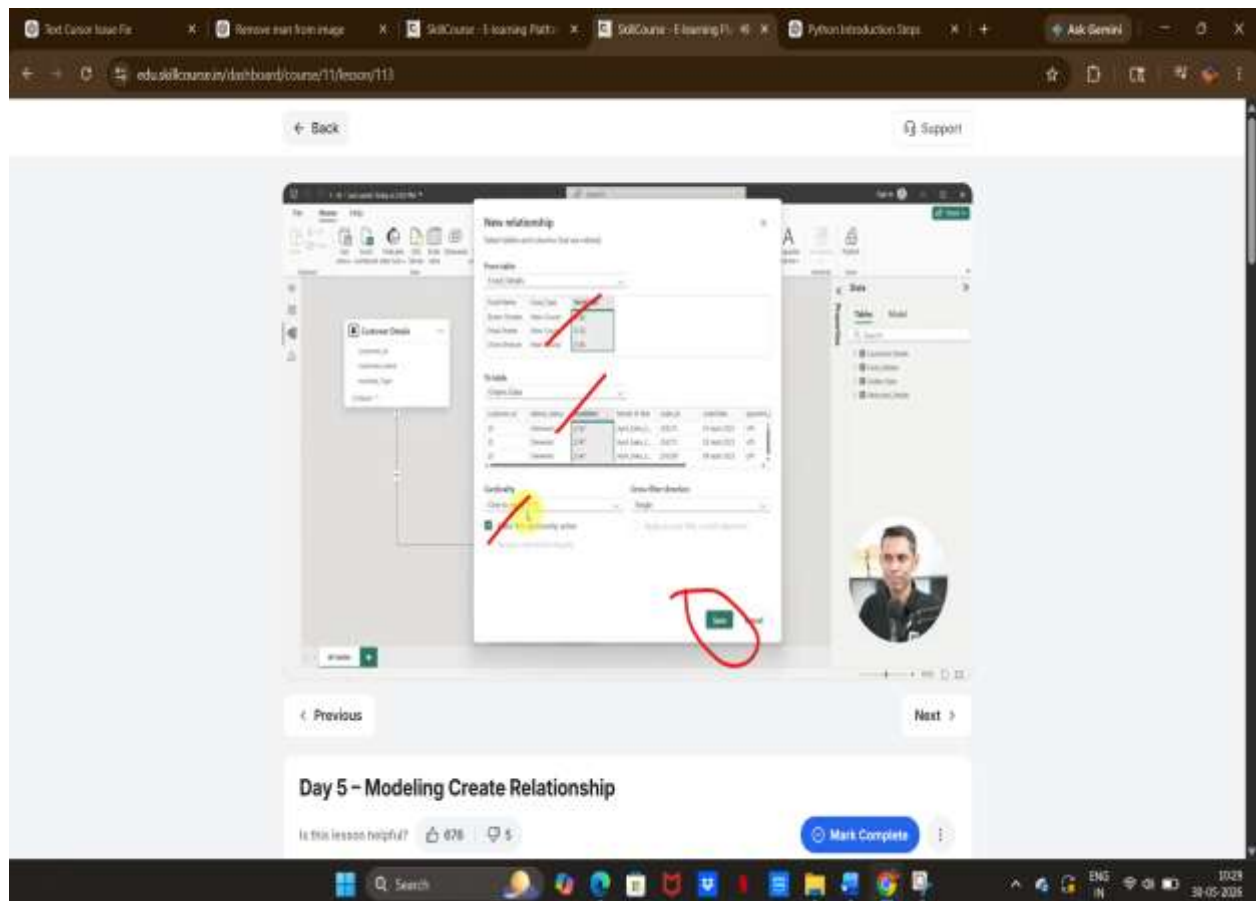
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Day 5 - Modeling Create Relationship

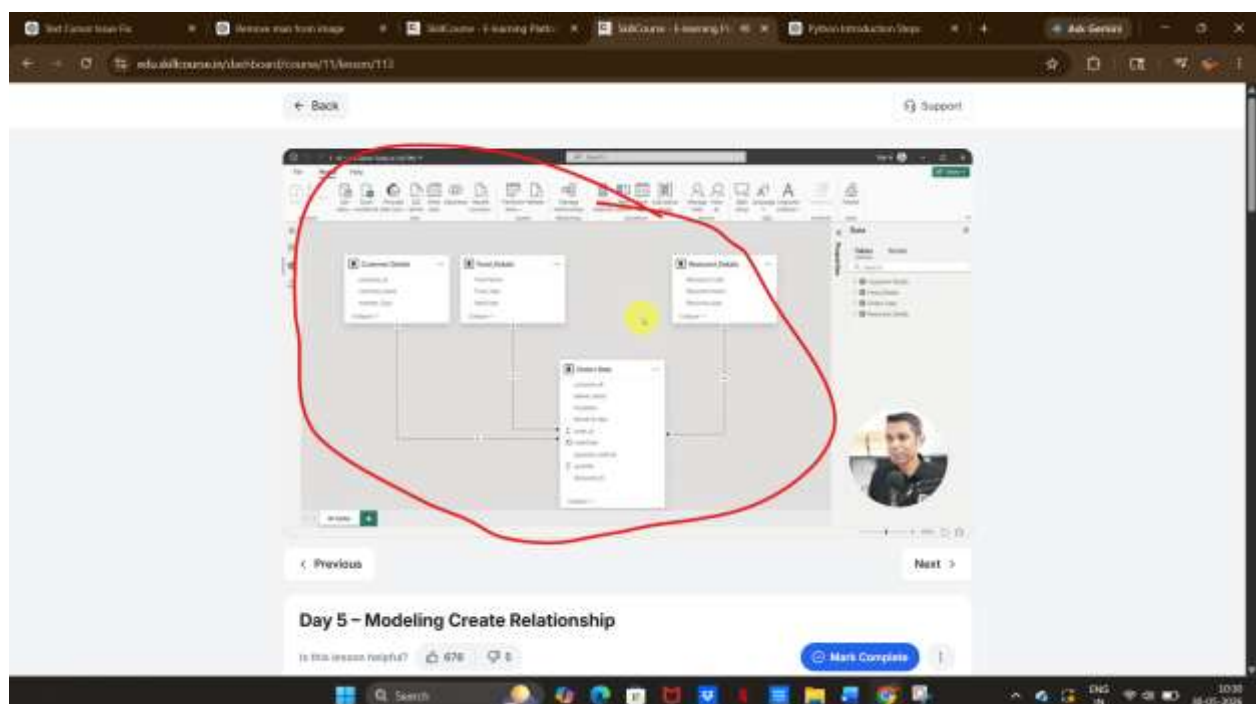
Is this lesson helpful? 676 5

Mark Complete

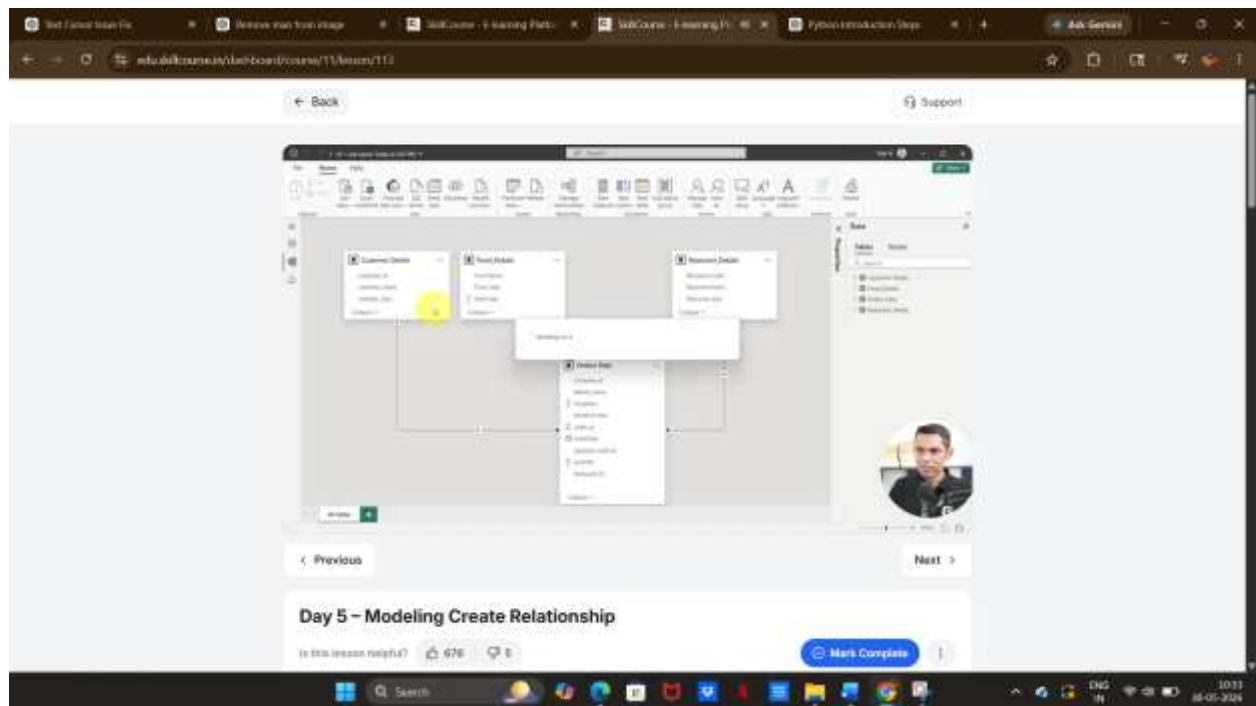
Windows Search



Step 3: so here we have final relationship btw tables.

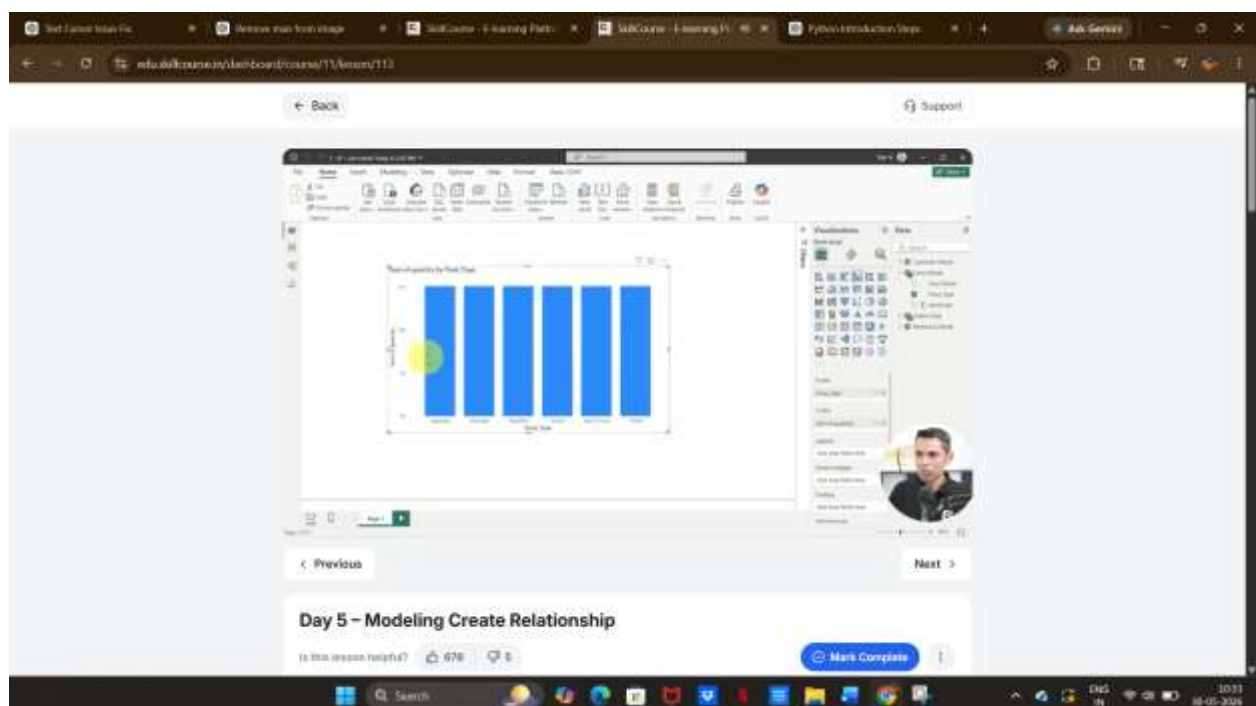


Step 4: if we have relationship we will get accurate result and charts .



Step 5: then we can see result is showing same in chart which is not accurate.

So relationship is very much important for accurate results and dashboards.



Interview Questions

Q1. What is Modeling?

Answer: Modeling is the process of creating relationships between tables in Power BI.

Q2. What is a Relationship?

Answer: A relationship is a connection between two tables using a common column.

Q3. What is a Primary Key?

Answer: A Primary Key contains unique values and is used on the One side of a relationship.

Q4. What is a Foreign Key?

Answer: A Foreign Key contains duplicate values and is used on the Many side of a relationship.

Q5. What is Cardinality?

Answer: Cardinality defines how tables are related to each other.

Q6. Which relationship is most commonly used?

Answer: One-to-Many (1:*).

Q7. What is Cross Filter Direction?

Answer: It decides which table can affect the other table when a filter is applied.

Q8. Difference between Active and Inactive Relationship?

Active Relationship

- Solid line
- Used automatically

Inactive Relationship

- Dotted line
- Not used automatically

Q9. What is a Fact Table?

Answer: A Fact Table contains numerical values such as Sales, Profit, and Quantity.

Q10. What is a Dimension Table?

Answer: A Dimension Table contains descriptive information such as Customer Name, Product Name, and Region.

Q11. What is Star Schema?

Answer: Star Schema is a data model where one Fact Table is connected to multiple Dimension Tables. It is the most recommended modeling structure in Power BI.

What is the extension of a Power BI Desktop file?

Answer:

The extension of a Power BI Desktop file is .pbix.

What does a .pbix file contain?

Answer:

A .pbix file contains data, relationships, Power Query transformations, DAX calculations, reports, and dashboard designs.

It also helps to keep data safe means if we make any changes it save and does not lose our work .